

APO-OLANZAPINE TABLETS

Olanzapine 5mg and 10mg Tablets

Antipsychotic Agent

PRODUCT DESCRIPTION

5 mg tablets: Each white, round biconvex film-coated tablet, engraved "APO" on one side and "OLA" over "5" on the other side. Thickness - 0.167 inch, Diameter - 0.3125 inch.

10 mg tablets: Each white, round biconvex film-coated tablet, engraved "APO" on one side and "OLA" over "10" on the other side. Thickness - 0.200 inch, Diameter - 0.4062 inch.

PHARMACOLOGY

Pharmacodynamics

Olanzapine, a thienobenzodiazepine, is an antipsychotic agent that demonstrates a broad pharmacologic profile across a number of receptor systems. Olanzapine displays high receptor affinity binding *in vitro* at dopamine D₁, D₃, D₄ (K_i = 11-31 nM), 5-HT_{2A/C} (K_i = 4 and 11 nM, respectively), 5-HT₃, 5-HT₆, muscarinic M₁-M₅ (K_i = 1.9-2.5 nM), adrenergic α₁ (K_i = 19 nM), and histamine H₁ (K_i = 7 nM) receptor subtypes, while displaying a lower affinity at dopamine D₁ and D₅ receptors (K_i = 51 – 119 nM).

Olanzapine also increases extracellular levels of dopamine in a regionally specific manner in the prefrontal cortex, similar to mood stabilizers, lithium and valproate.

Pharmacokinetics

Oral Administration:

Absorption: Olanzapine is well absorbed after oral administration, reaching peak plasma concentrations within 5 to 8 hours. The absorption is not affected by food.

Distribution: Plasma concentrations of orally administered olanzapine were linear and dose proportional. The maximum plasma concentrations (C_{max}) of olanzapine after single oral doses of 5, 10 and 15 mg averaged 7, 14, and 21 ng/mL, respectively (20 ng/mL = 0.064 μM). In young healthy adult, after once-a-day repeated dosing, steady-state C_{max} was approximately twice that achieved after a single dose (e.g. 23 ng/mL versus 12 ng/mL for a 10-mg dose). In the elderly, the steady state plasma concentration was approximately 3-fold higher than that achieved after a single dose (e.g. 16 ng/mL versus 5 ng/mL for a 5-mg dose). In both, young and elderly, steady-state concentrations of olanzapine were obtained after seven days of once daily dosing.

Over time and dosage range, pharmacokinetic parameters within an individual are very consistent. However, plasma concentrations, half-life and clearance of olanzapine may vary between individuals on the basis of smoking status, gender, and age (see Special Populations). Data from pooled, single dose pharmacokinetic studies showed the half-life of olanzapine to range from 21 to 54 hours (5th to 95th percentile), and the apparent plasma clearance to range from 12 to 47 L/hr (5th to 95th percentile).

The plasma protein binding of olanzapine was about 93% over the concentration range of about 7 to about 1000 ng/mL. Olanzapine is bound predominantly to albumin and α₁-acid glycoprotein.

Metabolism: Olanzapine is metabolized in the liver by conjugative and oxidative pathways. The

major circulating metabolite is the 10-N-glucuronide, which is pharmacologically inactive and does not pass the blood brain barrier. Cytochrome P450 isoforms CYP1A2 and CYP2D6 contribute to the formation of the N-desmethyl and 2-hydroxymethyl metabolites. The predominant pharmacologic activity is from the parent olanzapine.

In vitro microsomal studies show that olanzapine is a weak inhibitor of CYP1A2 ($K_i = 36 \mu\text{M}$), CYP2D6 ($K_i = 89 \mu\text{M}$), and CYP3A4 ($K_i = 490 \mu\text{M}$). Based upon these K_i values, little inhibition of these cytochrome P-450 enzymes is expected *in vivo* at concentrations below $5 \mu\text{M}$ (roughly 1500 ng/mL) because the olanzapine concentration will be less than 10% of its K_i value.

Olanzapine is thus not likely to cause clinically important pharmacokinetic drug-drug interactions mediated through the metabolic routes outlined above. (See DRUG INTERACTIONS section).

Elimination: After oral administration to healthy adult, the mean terminal elimination half-life was 33 hours (21 to 54 hours for 5th to 95th percentile) and the mean olanzapine plasma clearance was 26 L/hr (12 to 47 L/hr for the 5th to 95th percentile). Olanzapine pharmacokinetics varied on the basis of smoking status, gender, and age. Table 10 summarizes these effects:

Table 10: Olanzapine Key Pharmacokinetics

Patient Characteristics	Half-Life (hours)	Plasma Clearance (L/hr)
Nonsmoking	38.6	18.6
Smoking	30.4	27.7
Female	36.7	18.9
Male	32.3	27.3
Elderly (65 and older)	51.8	17.5
Nonelderly	33.8	18.2

Although smoking status, gender, and, to a lesser extent, age may affect olanzapine clearance and half-life, the magnitude of the impact of these single factors is small in comparison to the overall variability between individuals.

Special Populations and Conditions

Geriatrics: The mean elimination half-life of olanzapine was about 1.5 times greater in elderly (>65 years) than in non-elderly subjects (≤ 65 years). Caution should be used in dosing the elderly, especially if there are other factors that might additively influence drug metabolism and/or pharmacodynamic sensitivity (see **DOSAGE AND ADMINISTRATION** section).

Gender: Clearance of olanzapine is approximately 30% lower in women than in men. There were, however, no apparent differences between men and women in effectiveness or adverse effects. Dosage modifications based on gender should not be needed.

Race: In a study of Caucasians, Japanese, and Chinese subjects, there were no differences in olanzapine pharmacokinetics among the three populations. Cytochrome P450 isoform CYP2D6 status does not affect the metabolism of olanzapine.

Hepatic Insufficiency: No differences in the single-dose pharmacokinetics of oral olanzapine were noted in subjects with clinically significant cirrhosis (who were mostly smokers) when compared to healthy subjects (all non-smokers). Multiple-dose studies in patients with hepatic

impairment, however, have not been performed.

Renal Insufficiency: There was no significant difference in mean elimination half-life or olanzapine plasma clearance between subjects with severely impaired renal function compared to individuals with normal renal function. Approximately 57% of radio-labelled olanzapine is excreted in urine, principally as metabolites.

INDICATIONS AND CLINICAL USE

Olanzapine is indicated for the treatment of schizophrenia. Olanzapine is effective in maintaining the clinical improvement during continuing therapy in patients who have shown initial treatment response. Olanzapine is indicated for short-term treatment of acute manic episode associated with Bipolar I Disorder. Olanzapine is indicated for preventing recurrence of manic, mixed or depressive episodes in Bipolar I Disorder.

DOSAGE AND ADMINISTRATION

Schizophrenia: The recommended starting dose for olanzapine is 10 mg/day.

Manic episode: The starting dose is 15 mg as a single daily dose in monotherapy or 10 mg daily in combination therapy.

Preventing recurrence in bipolar disorder: The recommended starting dose is 10mg/day. For patients who have been receiving olanzapine for treatment of manic episode, continue therapy for preventing recurrence at the same dose. If a new manic, mixed, or depressive episode occurs, olanzapine treatment should be continued (with dose optimisation as needed), with supplementary therapy to treat mood symptoms, as clinically indicated. During treatment for schizophrenia, manic episode and recurrence prevention in bipolar disorder, daily dosage may subsequently be adjusted on the basis of individual clinical status within the range 5-20 mg/day. An increase to a dose greater than the recommended starting dose is advised only after appropriate clinical reassessment and should generally occur at intervals of not less than 24 hours.

Olanzapine can be given without regards for meals as absorption is not affected by food. Gradual tapering of the dose should be considered when discontinuing olanzapine.

Children: Olanzapine is not recommended for use in children and adolescents below 18 years of age due to a lack of data on safety and efficacy. A greater magnitude of weight gain, lipid and prolactin alterations has been reported in short term studies of adolescent patients than in studies of adult patients (see sections 4.4 and 4.8).

Elderly patients: A lower starting dose (5 mg/day) is not routinely indicated but should be considered for those 65 and over when clinical factors warrant.

Patients with renal and/or hepatic impairment: A lower starting dose (5 mg) should be considered for such patients. In cases of moderate hepatic insufficiency (cirrhosis, Child-Pugh Class A or B), the starting dose should be 5 mg and only increased with caution.

Gender: The starting dose and dose range need not be routinely altered for female patients relative to male patients.

Smokers: The starting dose and dose range need not be routinely altered for non-smoking patients relative to smoking patients.

When more than one factor is present which might result in slower metabolism (female gender, geriatric age, non-smoking status), consideration should be given to decreasing the starting

dose. Dose escalation, when indicated, should be conservative in such patients.

MODE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

APO-OLANZAPINE (olanzapine) is contraindicated in those patients with a known hypersensitivity to the drug or the excipients of the product and in patients with known risk for narrow-angle glaucoma.

WARNINGS AND PRECAUTIONS

During antipsychotic treatment, improvement in the patient's clinical condition may take several days to some weeks. Patients should be closely monitored during this period.

Dementia-related psychosis and/or behavioural disturbances

Olanzapine is not recommended for use in patients with dementia-related psychosis and/or behavioural disturbances because of an increase in mortality and the risk of cerebrovascular accident.

Parkinson's disease

The use of olanzapine in the treatment of dopamine agonist associated psychosis in patients with Parkinson's disease is not recommended.

Neuroleptic Malignant Syndrome (NMS)

NMS is a potentially life-threatening condition associated with antipsychotic medicinal products. Rare cases reported as NMS have also been received in association with olanzapine. Clinical manifestations of NMS are hyperpyrexia, muscle rigidity, altered mental status, and evidence of autonomic instability (irregular pulse or blood pressure, tachycardia, diaphoresis, and cardiac dysrhythmia). Additional signs may include elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acute renal failure. If a patient develops signs and symptoms indicative of NMS, or presents with unexplained high fever without additional clinical manifestations of NMS, all antipsychotic medicines, including olanzapine must be discontinued.

Hyperglycaemia and diabetes

Hyperglycaemia in some cases extreme and associated with ketoacidosis or hyperosmolar coma or death, has been reported in patients treated with atypical antipsychotics. Assessment of the relationship between atypical antipsychotics use and glucose abnormalities is complicated by the possibility of an increased background risk of diabetes mellitus in patients with schizophrenia and the increasing incidence of diabetes mellitus in the general population. Given this confounders, the relationship between atypical antipsychotic use and hyperglycaemia-related adverse events is not completely understood. However, epidemiological studies suggest an increased risk of treatment-emergent hyperglycaemia-related adverse events in patients treated with the atypical antipsychotics. Precise risk estimates for hyperglycaemia-related adverse events in patients treated with atypical antipsychotics are not available.

Patients with an established diagnosis of diabetes mellitus who are started on atypical antipsychotics should be monitored regularly for worsening of glucose control. Patients with risk factors for diabetes mellitus (e.g. obesity, family history of diabetes) who are starting treatment with atypical antipsychotics should undergo fasting blood glucose testing at the

beginning of treatment and periodically during treatment. Any patient treated with atypical antipsychotics should be monitored for symptoms of hyperglycaemia including polydipsia, polyuria, polyphagia, and weakness. Patients who develop symptoms of hyperglycaemia during treatment with atypical antipsychotics should undergo fasting blood glucose testing. In some cases, hyperglycaemia has resolved when the atypical antipsychotic was discontinued; however, some patients required continuation of anti-diabetic treatment despite discontinuation of the suspect drug.

Lipid alterations

Undesirable alterations in lipids have been observed. Lipid alterations should be managed as clinically appropriate, particularly in dyslipidemic patients and in patients with risk factors for the development of lipids disorders. Patients treated with any antipsychotic medicines, including olanzapine film-coated tablets, should be monitored regularly for lipids in accordance with utilised antipsychotic guidelines, e.g. at baseline, 12 weeks after starting olanzapine treatment and every 5 years thereafter.

Anticholinergic activity

As clinical experience with olanzapine in patients with concomitant illness is limited, caution is advised when prescribing for patients with prostatic hypertrophy, or paralytic ileus and related conditions.

Hepatic function

Transient, asymptomatic elevations of hepatic aminotransferases, ALT, AST have been seen commonly, especially in early treatment. Caution should be exercised and follow-up organised in patients with elevated ALT and/or AST, in patients with signs and symptoms of hepatic impairment, in patients with pre-existing conditions associated with limited hepatic functional reserve, and in patients who are being treated with potentially hepatotoxic medicines. In cases where hepatitis (including hepatocellular, cholestatic or mixed liver injury) has been diagnosed, olanzapine treatment should be discontinued.

Neutropenia

Caution should be exercised in patients with low leukocyte and/or neutrophil counts for any reason, in patients receiving medicines known to cause neutropenia, in patients with a history of drug-induced bone marrow depression/toxicity, in patients with bone marrow depression caused by concomitant illness, radiation therapy or chemotherapy and in patients with hypereosinophilic conditions or with myeloproliferative disease. Neutropenia has been reported commonly when olanzapine and valproate are used concomitantly.

Discontinuation of treatment

Acute symptoms such as sweating, insomnia, tremor, anxiety, nausea, or vomiting have been reported rarely when olanzapine is stopped abruptly.

QT interval

Caution should be exercised when olanzapine is prescribed with medicines known to increase QTc interval, especially in the elderly, in patients with congenital long QT syndrome, congestive heart failure, heart hypertrophy, hypokalaemia or hypomagnesaemia.

Thromboembolism

Temporal association of olanzapine treatment and venous thromboembolism has been reported uncommonly. A causal relationship between the occurrence of venous thromboembolism and treatment with olanzapine has not been established. However, since

patients with schizophrenia often present with acquired risk factors for venous thromboembolism all possible risk factors of VTE e.g. immobilisation of patients, should be identified and preventive measures undertaken.

General CNS activity

Given the primary CNS effects of olanzapine, caution should be used when it is taken in combination with other centrally acting medicines and alcohol. As it exhibits *in vitro* dopamine antagonism, olanzapine may antagonize the effects of direct and indirect dopamine agonists.

Seizures

Olanzapine should be used cautiously in patients who have a history of seizures or are subject to factors which may lower the seizure threshold. Seizures have been reported to occur uncommonly in patients when treated with olanzapine. In most of these cases, a history of seizures or risk factors for seizures was reported.

Tardive Dyskinesia

Olanzapine was associated with a statistically significant lower incidence of treatment emergent dyskinesia. However the risk of tardive dyskinesia increases with long term exposure, and therefore if signs or symptoms of tardive dyskinesia appear in a patient on olanzapine, a dose reduction or discontinuation should be considered. These symptoms can temporally deteriorate or even arise after discontinuation of treatment.

Postural hypotension

Postural hypotension was infrequently observed in elderly. It is recommended that blood pressure is measured periodically in patients over 65 years.

Sudden cardiac death

In post-marketing reports with olanzapine, the event of sudden cardiac death has been reported in patients with olanzapine.

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)

Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS) has been reported with olanzapine exposure. DRESS consists of a combination of three or more of the following: cutaneous reaction (such as rash or exfoliative dermatitis), eosinophilia, fever, lymphadenopathy and one or more systemic complications such as hepatitis, nephritis, pneumonitis, myocarditis, and pericarditis. Discontinue olanzapine if DRESS is suspected.

Paediatric population

Olanzapine is not indicated for use in the treatment of children and adolescents.

Effects on ability to drive and use machines

Olanzapine may cause somnolence and dizziness, patients should be cautioned about operating machinery, including motor vehicles.

DRUG INTERACTIONS

Paediatric population

Interaction studies have only been performed in adults.

Potential interactions affecting olanzapine

Since olanzapine is metabolised by CYP1A2, substances that can specifically induce or inhibit this isoenzyme may affect the pharmacokinetics of olanzapine.

Induction of CYP1A2

The metabolism of olanzapine may be induced by smoking and carbamazepine, which may lead

to reduced olanzapine concentrations. Only slight to moderate increase in olanzapine clearance has been observed. The clinical consequences are likely to be limited, but clinical monitoring is recommended and an increase of olanzapine dose may be considered if necessary.

Inhibition of CYP1A2

Fluvoxamine, a specific CYP1A2 inhibitor, has been shown to significantly inhibit the metabolism of olanzapine. A lower starting dose of olanzapine should be considered in patients who are using fluvoxamine or any other CYP1A2 inhibitors, such as ciprofloxacin. A decrease in the dose of olanzapine should be considered if treatment with an inhibitor of CYP1A2 is initiated.

Decreased bioavailability

Activated charcoal reduces the bioavailability of oral olanzapine by 50 to 60% and should be taken at least 2 hours before or after olanzapine.

Fluoxetine (a CYP2D6 inhibitor), single doses of antacid (aluminium, magnesium) or cimetidine have not been found to significantly affect the pharmacokinetics of olanzapine.

Potential for olanzapine to affect other medicinal products

Olanzapine may antagonise the effects of direct and indirect dopamine agonists.

Olanzapine does not inhibit the main CYP450 isoenzymes *in vitro* (e.g. 1A2, 2D6, 2C9, 2C19, 3A4). Thus no particular interaction is expected as verified through *in vivo* studies where no inhibition of metabolism of the following active substances was found: tricyclic antidepressant (representing mostly CYP2D6 pathway), warfarin (CYP2C9), theophylline (CYP1A2) or diazepam (CYP3A4 and 2C19).

Olanzapine showed no interaction when co-administered with lithium or biperiden.

Therapeutic monitoring of valproate plasma levels did not indicate that valproate dosage adjustment is required after the introduction of concomitant olanzapine.

General CNS activity

Caution should be exercised in patients who consume alcohol or receive medicinal products that can cause central nervous system depression.

The concomitant use of olanzapine with anti-Parkinsonian medicinal products in patients with Parkinson's disease and dementia is not recommended.

QTc interval

Caution should be used if olanzapine is being administered concomitantly with medicinal products known to increase QTc interval.

PREGNANCY AND LACTATION

Pregnant Women:

There are no adequate and well-controlled studies in pregnant women. Patients should be advised to notify their physician if they become pregnant or intend to become pregnant during treatment with olanzapine. Because human experience in pregnant females is limited, this drug should be used in pregnancy only if the potential benefit justifies the potential risk to the fetus.

Non-Teratogenic Effects

Neonates exposed to antipsychotic drugs (including olanzapine) during the third trimester of pregnancy are at risk for extrapyramidal and/or withdrawal symptoms following delivery. There have been reports of agitation, hypertonia, hypotonia, tremor, somnolence, respiratory distress

and feeding disorder in these neonates. These complications have varied in severity; while in some cases symptoms have been self-limited, in other cases neonates have required intensive care unit support and prolonged hospitalization.

Olanzapine should not be used during pregnancy unless the expected benefits to the mother markedly outweigh the potential risks to the fetus.

Labour and Delivery:

Parturition in rats was not affected by olanzapine. The effect of olanzapine on labour and delivery in humans is not known.

Nursing Women:

Lactation:

In a study in lactating, healthy women, olanzapine was excreted in breast milk. Mean infant exposure (mg/kg) at steady state was estimated to be 1.8% of the maternal olanzapine dose (mg/kg). Patients should be advised not to breast feed an infant if they are taking olanzapine.

ADVERSE REACTIONS

Adults

The most frequently reported adverse reactions associated with the use of olanzapine were somnolence, weight gain, eosinophilia, elevated prolactin, cholesterol, glucose and triglyceride levels, glucosuria, increased appetite, dizziness, akathisia, parkinsonism, leukopenia, neutropenia, dyskinesia, orthostatic hypotension, anticholinergic effects, transient asymptomatic elevations of hepatic aminotransferases, rash, asthenia, fatigue, pyrexia, arthralgia, increased alkaline phosphatase, high gamma glutamyltransferase, high uric acid, high creatine phosphokinase and oedema.

Tabulated list of adverse reactions

Very common	Common	Uncommon	Rare	Not known
Blood and the lymphatic system disorders				
	Eosinophilia Leukopenia Neutropenia		Thrombocytopenia	
Immune system disorders				
		Hypersensitivity		
Metabolism and nutrition disorders				
Weight gain	Elevated	Development or exacerbation of	Hypothermia	

	cholesterol levels Elevated glucose levels Elevated triglyceride levels Glucosuria Increased appetite	diabetes occasionally associated with ketoacidosis or coma, including some fatal cases		
Nervous system disorders				
Somnolence	Dizziness Akathisia Parkinsonism Dyskinesia	Seizures where in most cases a history of seizures or risk factors for seizures were reported Dystonia (including oculogyration) Tardive dyskinesia Amnesia Dysarthria Restless legs syndrome	Neuroleptic malignant syndrome Discontinuation symptoms	
Cardiac disorders				
		Bradycardia QT _c prolongation	Ventricular tachycardia/fibrillation, sudden death	
Vascular disorders				
Orthostatic hypotension		Venous thromboembolism (including pulmonary embolism and deep vein thrombosis)		
Respiratory, thoracic and mediastinal disorders				
		Epistaxis		

Gastrointestinal disorders				
	Mild, transient anticholinergic effects including constipation and dry mouth	Abdominal distension	Pancreatitis	
Hepato-biliary disorders				
	Transient, asymptomatic elevations of hepatic aminotransferases (ALT, AST), especially in early treatment		Hepatitis (including hepatocellular, cholestatic or mixed liver injury)	
Skin and subcutaneous tissue disorders				
	Rash	Photosensitivity reaction Alopecia		Drug Reaction with Eosinophilia and Systemic Symptoms (DRESS)
Musculoskeletal and connective tissue disorders				
	Arthralgia		Rhabdomyolysis	
Renal and urinary disorders				
		Urinary incontinence, urinary retention Urinary hesitation		
Pregnancy, puerperium and perinatal conditions				
				Drug withdrawal syndrome neonatal

Reproductive system and breast disorders				
	Erectile dysfunction in males Decreased libido in males and females	Amenorrhea Breast enlargement Galactorrhea in females Gynaecomastia/breast enlargement in males	Priapism	
General disorders and administration site conditions				
	Asthenia Fatigue Oedema Pyrexia			
Investigations				
Elevated plasma prolactin levels	Increased alkaline phosphatase High creatine phosphokinase High Gamma glutamyltransferase High uric acid	Increased total bilirubin		

Long-term exposure (at least 48 weeks)

The proportion of patients who had adverse, clinically significant changes in weight gain, glucose, total/LDL/HDL cholesterol or triglycerides increased over time. In adult patients who completed 9-12 months of therapy, the rate of increase in mean blood glucose slowed after approximately 6 months.

Additional information on special populations

In elderly patients with dementia, olanzapine treatment was associated with a higher incidence of death and cerebrovascular adverse reactions.

Very common adverse reactions associated with the use of olanzapine in this patient group were abnormal gait and falls. Pneumonia, increased body temperature, lethargy, erythema, visual hallucinations and urinary incontinence were observed commonly.

In patients with drug-induced (dopamine agonist) psychosis associated with Parkinson's disease, worsening of Parkinsonian symptomatology and hallucinations were reported very commonly.

In patients with bipolar mania, valproate combination therapy with olanzapine resulted in an incidence of neutropenia; a potential contributing factor could be high plasma valproate levels. Olanzapine administered with lithium or valproate resulted in increased levels of tremor, dry mouth, increased appetite, and weight gain. Speech disorder was also reported commonly. During treatment with olanzapine in combination with lithium or divalproex, an increase from baseline body weight occurred in patients during acute treatment (up to 6 weeks). Long-term olanzapine treatment (up to 12 months) for recurrence prevention in patients with bipolar disorder was associated with an increase from baseline body weight.

Paediatric population

Olanzapine is not indicated for the treatment of children and adolescent patients below 18 years.

The following table summarises the adverse reactions reported with a greater frequency in adolescent patients (aged 13-17 years) than in adult patients or adverse reactions only identified during short-term clinical trials in adolescent patients. Clinically significant weight gain appears to occur more frequently in the adolescent population compared to adults with comparable exposures. The magnitude of weight gain and the proportion of adolescent patients who had clinically significant weight gain were greater with long-term exposure (at least 24 weeks) than with short-term exposure.

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Metabolism and nutrition disorders

Very common: Weight gain, elevated triglyceride levels, increased appetite.

Common: Elevated cholesterol levels

Nervous system disorders

Very common: Sedation (including: hypersomnia, lethargy, somnolence).

Gastrointestinal disorders

Common: Dry mouth

Hepato-biliary disorders

Very common: Elevations of hepatic aminotransferases (ALT/AST).

Investigations

Very common: Decreased total bilirubin, increased GGT, elevated plasma prolactin levels.

OVERDOSAGE

Signs and Symptoms

Very common symptoms reported in olanzapine overdose ($\geq 10\%$ incidence) include tachycardia, agitation/aggressiveness, dysarthria, various extrapyramidal symptoms, and reduced level of consciousness ranging from sedation to coma.

Other medically significant sequelae of olanzapine overdose include delirium, convulsion, possible neuroleptic malignant syndrome, respiratory depression, aspiration, hypertension or

hypotension, cardiac arrhythmias (<2% of overdose cases) and cardiopulmonary arrest. Fatal outcomes have been reported for acute overdoses as low as 450 mg of oral olanzapine but survival has also been reported following acute overdose of approximately 2,000 mg of oral olanzapine.

Management of Overdose

There is no specific antidote for olanzapine. Induction of emesis is not recommended. Standard procedures for management of overdose may be indicated (i.e. gastric lavage, administration of activated charcoal). The concomitant administration of activated charcoal was shown to reduce the oral bioavailability of olanzapine by 50 to 60%.

Symptomatic treatment and monitoring of vital organ function should be instituted according to clinical presentation, including treatment of hypotension and circulatory collapse and support of respiratory function. Do not use epinephrine, dopamine, or other sympathomimetic agents with beta-agonist activity since beta stimulation may worsen hypotension.

Cardiovascular monitoring is necessary to detect possible arrhythmias. Close medical supervision and monitoring should continue until the patient recovers.

STORAGE CONDITIONS

HDPE Bottle: Store below 25°C. Protect from light and moisture.

Blister in box: Store below 30°C. Protect from light and moisture.

AVAILABILITY OF DOSAGE FORMS

Apo-Olanzapine is available in 5mg tablet and 10mg tablet.

AVAILABLE PACK SIZE

5mg - Box of 28 tablets.

5mg - Bottle of 100 tablets.

10mg - Box of 28 tablets.

10mg - Bottle of 60 tablets.

Not all pack sizes are marketed.

MANUFACTURER

Apotex Inc.

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DATE OF REVISION

18 Apr 2024